

TANKLESS OPTION IMPROVING

If cost and installation barriers can be overcome, tankless water heaters will likely replace the storage water heater that for over 50 years has been a part of American culture.

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Storage gas water heaters are the predominant water-heating appliances used in homes in the United States. They have a low first cost, and their hot water delivery meets the expectations of most U.S. homeowners. But storage water heaters are a technology that has not changed significantly over the past thirty years. Instantaneous, or tankless, gas water heaters on the other hand have changed in recent years—for the better. The newer tankless water heaters now offer significant improvements over storage tank units.

Variable burner capacity, higher heating capacities, and sophisticated controls have markedly improved the performance and hot water delivery characteristics of tankless units. The elimination of standing pilot lights has led to a noticeable increase in the efficiency of tankless units; Energy Factors (EFs) for tankless heaters are more than 30% greater than EFs for typical storage gas water heaters. Efficiency isn't the only means in which these units have a very clear advantage over storage-type heaters: Tankless water heaters can provide a limitless supply of hot water; they can be more maintainable than storage water heaters; and their small size frees up floor space for other uses.

The increased efficiency of tankless water heaters can provide substantial natural gas savings and reduced carbon emissions. For this reason, the Pacific Gas and Electric Company (PG&E)

asked our company, Davis Energy Group, to study tankless gas water heater performance and to identify barriers to wider acceptance of this technology. The utility wanted to gain a better understanding of the application of tankless technology in single-family and multifamily new construction and retrofit markets. Typical California water heating energy use is roughly equivalent to space heating consumption. A 30% reduction in water heating energy use, which is achievable with tankless water heaters, would reduce California's total natural gas consumption by an astound-

ing 2.5%, and the reduction in gas consumption would eliminate 3.3 million tons of carbon emissions annually. Given the number of residential gas water heaters in California, the potential for saving energy is substantial (see Table 1).

State of the Art

Tankless gas water heaters employ a burner and a heat exchanger that contains a small volume of water (typically less than 1 gallon). Water is heated in a single pass through the heat exchanger. Varying burner output proportional to



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the flow rate regulates supply water temperature. Newer types include electronic (spark) ignition, greater modulation of the firing rate, and more intelligent controls—features that represent a significant improvement over the models from ten to twenty years ago. Many tankless water heaters utilize combustion air blowers to achieve higher output and efficiency, and to

homes. Their high capacity ensures that homeowners will not run out of hot water under normal usage patterns.

Tankless water heater models exhibit a range of performance characteristics (see Table 2). The models listed in the table have the capability to modulate the burner-firing rate down as little as 8% of their maximum heating capacity. This is an important feature, which allows water

losses. As the hot water load decreases, the standby losses become a larger percentage of the total usage. Tankless water heaters are minimally affected by load, but their efficiency is impacted by the number of times they cycle in response to hot water draws. Maximum efficiency is achieved when hot water draws are tightly clustered or continuous, rather than dispersed throughout the day. Currently, DOE applies the same test procedure to both storage and tankless water heaters. Thus, the current minimum EF standard for tankless heaters is 0.67, a higher standard than is required for storage-type heaters but lower than the EF for any tankless product currently on the market. DOE may be considering development of a separate standard for tankless water heaters in 2006.

The long-term reliability of the new breed of tankless water heaters is uncertain, given their recent introduction to the market. Most tankless models have a life expectancy of more than 20 years, and have easily replaceable parts that can extend their life by many more years. To ensure continued performance, manufacturers recommend that owners periodically flush the heat exchanger with a mild acid solution to remove scale from the heat exchanger.

As well as requiring a minimum flow rate, tankless heaters require about five seconds to bring cold water up to temperature. Homeowners may notice these shortcomings, but our experience with two field demonstrations suggests that most homeowners either are not aware of these shortcomings or readily adapt to them.

Extensive literature searches failed to produce accurate data on the relative market share of tankless water heaters versus conventional residential water heaters. But sales managers for the leading manufacturers estimate market share at roughly 1%–2%.

Efficiency Advantage and Potential Energy Savings

Current National Appliance Energy

Table 1. California Gas Storage Water Heater Inventory

Utility Service Area	Existing Housing Stock	Number of Gas Water Heaters	Gas Water Heater Saturation
Pacific Gas and Electric	4,628,766	3,813,231	82%
Sacramento Municipal Utility	487,468	366,493	75%
Southern California Edison	4,247,004	3,550,431	84%
LA Dept of Water & Power	1,300,720	1,142,707	88%
San Diego Gas and Electric	1,156,391	955,586	83%
Burbank, Glendale, Pasadena	169,700	147,743	87%
Statewide Total	11,990,049	9,976,191	83%

Source: California Energy Demand 2003–2013 Forecast. Sacramento, California Energy Commission, August 2003.

Table 2. Instantaneous Water Heater Performance Characteristics

Manufacturer	Model Number	EF	Max Input (Btu/h)	Min Input (Btu/h)
Bosch/Aquastar	AQ125FX	0.78	125,000	35,000
	AQ240FX	0.81	165,000	37,000
Noritz America	N-063	0.81	185,000	19,000
	N-084	0.80	236,000	21,000
Rinnai	Continuum 2402	0.82	180,000	20,000
	Continuum 2424	0.82	180,000	19,000
Takagi	FLASH/T-K2	0.85	185,000	20,000
	FLASH/T-KJr.	0.83	140,000	19,500

allow horizontal direct venting. Lower-cost models use a push-button piezo-electric igniter to start the pilot, use the pilot to ignite the main burner, and are naturally vented. Maximum gas input rates range from about 60,000 to 240,000 Btu per hour. Lower-capacity units are intended for point-of-use, or small-load applications, while larger units can be used in combined hydronic applications. Most manufacturers supply both natural gas and propane-fueled units.

Several manufacturers now produce tankless gas units that are capturing a larger market share than previous models because they respond quickly to changing loads—with little or no change in output temperature—and meet the hot water needs of most single-family

to be heated at flow rates as low as 0.5 gallons per minute (gpm). At flow rates less than 0.5 gpm, most units will not fire. Models with capacities lower than 120,000 Btu per hour are marginal for whole-house applications with simultaneous hot water draws (for example, two showers or a shower coincident with clothes washing). Models with capacities higher than 160,000 Btu per hour are better suited for whole-house applications and the replacement of tank-type units.

The EFs listed in Table 2 are based on DOE test procedures that calculate total water heater energy use and energy delivered at a standard hot water load of 64.3 gallons per day. For gas storage water heaters, a significant fraction of the energy use is consumed due to standby

Conservation Act (NAECA) requirements specify minimum EFs for 40- and 50-gallon gas water heaters of 0.59 and 0.58, respectively. For typical California new construction, water heaters with an EF rating of 0.60 are commonly installed as an expedient means of complying with Title 24, the state's energy code.

In existing housing stock, the average EF is probably closer to the NAECA standard in effect from 1992 to 2003, which for a 40-gallon water heater was 0.54. However, given that the best opportunity to replace storage with tankless heaters occurs when the owner is contemplating heater replacement, the baseline should be the current minimum standard, or 0.59 for a typical 40-gallon unit. The EF for modern tankless units without pilots ranges from 0.78 to 0.84. We selected a median value of 0.82 for this analysis.

Since efficiency varies with load, using the rated EF to estimate savings for water heaters that produce more or less than the DOE-standard 64.3 gallons per day will give inaccurate results. The California Title 24 energy standards apply a "load-dependent EF" which adjusts the rated EF, based on hot water usage, to calculate water-heating energy use. A similar adjustment is needed to develop reasonably accurate estimates of tankless water heater energy savings.

Davis Energy Group, with the support of DOE's Building America program, completed a field test of the relationship between load and efficiency for both types of water heater. In this test, our group monitored a storage water heater (flow rate, temperatures, gas use) in a single-family residence for 29 days, and then replaced it with a tankless water heater that was monitored for an additional 19 days. The storage water heater was about four years old and had a 0.62 EF. The replacement tankless heater had an EF of 0.82.

Plotting measured efficiency from this test (heat supplied divided by gas energy consumption) versus the daily volume of hot water drawn yields some interesting conclusions (see Figure 1). (The solid lines in Figure 1 are regression fits to the measured data points.) The relationship between hot water use and efficiency (as

a result of the increased impact of standby loss) is clearly demonstrated by the storage water heater curve. Extrapolation of this curve for the storage water heater was necessary since monitored daily loads never exceed 40 gallons per

The wider scatter of points for the tankless heater shows that there is not as direct a correlation between efficiency and hot water use as there is for the storage water heater. Water left in the tankless heater at the end of a cycle is

Table 3. Projected Performance as a Function of Water Heating Load

Daily Load (Gal/Day)	Calculated LDEF		Annual Use (Therms)		Annual Savings (Therms)
	Storage/Gas	Tankless	Storage/Gas	Tankless	
10	0.24	0.68	95	34	61
20	0.36	0.70	125	65	60
30	0.41	0.71	166	96	70
40	0.42	0.73	215	125	90
50	0.45	0.75	256	153	103
60	0.52	0.76	265	179	86
36.3 MF	0.42	0.72	197	115	82
53.2 SF	0.46	0.75	263	162	102

Field Measurements of Gas Storage and Tankless Water Heater Performance

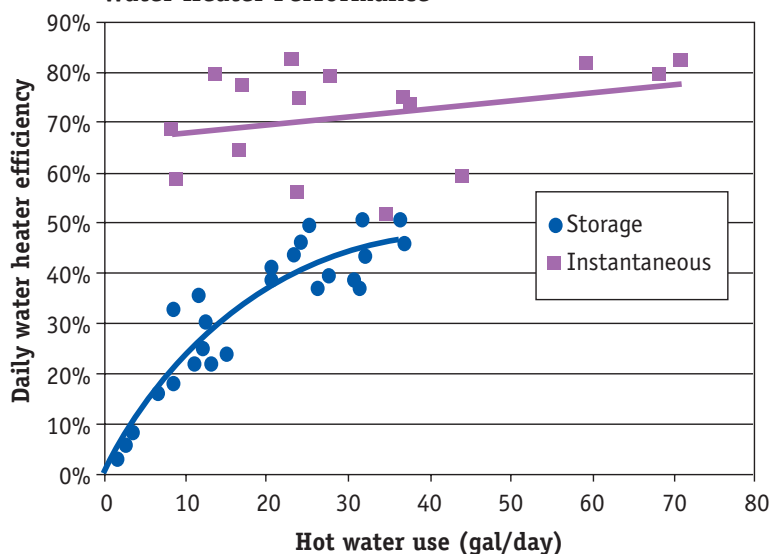


Figure 1. The relationship between hot water use and efficiency (as a result of the increased impact of standby loss) is clearly demonstrated by the storage water heater curve. The wider scatter of points for the tankless heater shows that there is not as direct a correlation between efficiency and hot water use as there is for the storage water heater.

day. Extrapolating the data to the 64.3-gallon-per-day rating point yields an efficiency that is below the nominal 0.62 EF. Diminished efficiency due to tank scale accumulation, hot water use patterns, and other differences between DOE and field test methods may account for this difference.

equivalent to standby losses for a storage water heater, albeit on a much smaller scale. The upward trend of the tankless efficiency curve is due to the fact that as daily hot water use increases, individual draws are likely to be larger, and large draws correlate with improved efficiency due to less cycling. Each time the

tankless heater cycles, the heat exchanger and small volume of contained water must be brought up to temperature regardless of the magnitude of the hot water draw. Tankless heater losses go to zero between draws, unlike storage water heaters that demonstrate constant standby loss.

The regression curves shown in Figure 1 were used to calculate load-dependent EF for each water heater type over a range of daily hot water use volumes and these were then used to calculate energy use and savings. The California Title 24 method for calculating water heater energy use correlates hot water use with floor area (smaller houses have smaller loads). This correlation was used to determine representative use quantities for typical single- and multifamily units.

We assumed average conditioned floor areas of 2,348 ft² and 1,127 ft², based on current new construction data, for the single and multifamily units, resulting in daily hot water use quantities of 53.2 and 36.3 gallons per day, respectively. The performance curves were used to generate savings projections for tankless versus storage type water heaters. Results indicate that energy savings increase with increasing load and peak at approximately 50 gallons per day (see Table 3).

The single- and multifamily energy use and savings values are listed in the last two rows of Table 3, and indicate annual projected savings of 82 and 102 therms for multifamily (MF) and single-family (SF) residential building types, respectively. Two factors contribute to make these savings estimates conservative. First, storage water heater gas consumption is based on the monitored performance of a water heater with a 0.62 EF instead of the baseline value of 0.60. Second, field test data on which estimates are based were collected during the month of August, when ambient temperatures were above the seasonal average. As a result, the standby loss from the storage water heater is lower than the seasonal value. (Tankless water heater performance is relatively independent of ambient temperature because these units store a minimal amount of water.)

We calculated homeowner cost savings based on an average gas rate of \$0.91 per therm (the average PG&E retail rate over the past four years). Projected annual savings under these conservative assumptions—gas prices have already risen higher than our projections—range from \$75 for multifamily homes (82 therms per year) to \$93 for single-family homes (102 therms per year). One way to evaluate economic value in new homes is to compare the amortized cost of the energy measure to the energy savings. For single-family new construction, the estimated \$1,100 incre-

• Cost is the single greatest barrier to the purchase of tankless water heaters; this was mentioned by everyone that we interviewed.

• Because of their higher gas input rating, tankless heaters require 3/4-inch or 1-inch gas piping, depending on the distance from the meter to the heater, compared to the 1/2-inch pipe size required in most cases for tank-type water heaters. The incremental cost of installing larger-diameter gas pipe for new construction is minimal. For retrofit applications, the existing line must usually be replaced, and this cost may exceed the cost of the tankless heater.

• Indoor installations require a Category 3 stainless steel exhaust vent. This code requirement contributes to the cost barrier. Minimal vent piping is required for indoor units vented through an exterior wall, and most outdoor units do not require venting.

• Unlike gas-fired storage water heaters, most tankless units require 120V power. The cost impact in new construction is minimal. For retrofit applications, the cost will vary depending on whether there is an existing outlet within 6 ft of the heater (the maximum length allowed for appliance cords). One model uses a patented “hydropower ignition” technology to eliminate the need for electrical connection. With this approach, flowing water drives a turbine to create the spark ignition.

• Most tankless water heaters with electronic ignition will not operate during a power outage. At first, homeowners may not see this as a barrier, but if enough disgruntled homeowners lose hot water during a power failure, the market could suffer.

• Manufacturers recommend periodic maintenance to remove water deposits from the heat exchanger. This involves isolating the water heater from the hot and cold water system and flushing the heat exchanger with a mild acid solution. The frequency of maintenance depends upon the local water

Table 4. Projected Savings Based on Load and Gas Rate Variations

Daily Load (Gal/Day)	Average Rate (per Therm)	Projected Savings (Therms/Year)	Annual Savings
20	\$0.70	60	\$ 42
20	\$1.00	60	\$ 60
20	\$1.30	60	\$ 78
40	\$0.70	90	\$ 63
40	\$1.00	90	\$ 90
40	\$1.30	90	\$117
60	\$0.70	86	\$ 60
60	\$1.00	86	\$ 86
60	\$1.30	86	\$112

mental cost translates to a monthly increase in mortgage cost that is less than the projected gas savings (\$7.75 monthly savings versus \$6.60 monthly mortgage cost). Multifamily new-construction economics are slightly less favorable due to the lower projected savings (\$6.25 per month). Table 4 demonstrates how operating cost savings are projected to vary with load magnitude and average gas rate. Under a best-case costing scenario, retrofit applications demonstrate simple paybacks ranging from 12 to 15 years.

Barriers to Acceptance

We compiled the following list of barriers to adopting tankless water heaters from interviews with manufacturers, contractors, and building officials. Some of the identified barriers apply only to retrofit installations, while some apply to both new and retrofit installations. Barriers are listed based on our perceived ranking, from most significant to least significant:

quality; in areas with extremely hard water, flushing may be required as frequently as once every year. The cost-per-service interval ranges from near zero (for do-it-yourselfers) to about \$200. If not flushed, tankless heaters may fail in five to seven years. Some units include a diagnostic output that alerts the homeowner when the heat exchanger needs to be flushed.

- Specialized training is required for proper installation and servicing of tankless heaters. No such training is required for gas storage water heaters. While manufacturers offer training classes, contractors have little incentive to attend if they seldom encounter tankless heaters in the field.

- Homeowners and contractors who have no experience with tankless water heaters are less likely to have confidence in their performance and durability.

- Tankless water heaters require 5–15 seconds to heat water to the set temperature. If there are already long waiting times for hot water delivery to remote fixtures because of poor piping design, the added waiting time could make delivery times unacceptable (see “Hot Water Runs Cold,” *HE* March/April '05, p. 28). This problem is more significant in retrofit applications, where owners are accustomed to the waiting time of their storage water heater.

- One vendor noted that the building department in his area does not allow tankless heaters. While there is nothing in the plumbing codes to prohibit listed tankless heaters, the fact that building officials may not understand their use is still a barrier. The building officials interviewed for this study were either neutral or somewhat negative on the value of tankless heaters, but they did not indicate that they would not allow them.

- Tankless water heaters are not recommended for installation in homes that have galvanized steel piping unless a filter is added to the water heater inlet pipe. Flakes of scale from the piping can lodge in the screen and/or the flow meter and cause malfunctions. Galvanized piping is found in homes built in 1960 or earlier.

- Tankless water heaters can be purchased from big-box retailers by owner-installers, but vent kits are offered only through wholesalers. This means that owners cannot install preventing indoor units without the help of a contractor.

- Because they lack water storage that serves as a buffer to hot water use, tankless water heaters require more careful sizing than storage-type heaters. Undersizing has led to inadequate hot water delivery when multiple fixtures are in use, and has generated a small population of dissatisfied users. This barrier can be overcome by education.

- Some of the building professionals interviewed felt that the largest market for tankless water heaters was in luxury homes, where unlimited hot water volume is more important than reducing water use or energy use. If this is true, then the unlimited supply of hot water could result in diminished overall energy savings stemming from longer showers or increased use of Jacuzzi tubs.

- When tankless water heaters are used with hot water recirculation systems that employ motion sensors or push buttons to activate the recirculation pump, the pump may shut off before the hot water draw begins. Because of the start-up delay of the water heater, the user may experience a stream of hot water followed by a brief stream of cold water (while the heater is firing), followed again by hot water. Educating the owner on the proper timing for activating the pump can prevent dissatisfaction with the tankless heater.

- Tankless water heaters require a minimum flow rate, (usually about 0.5 gpm) to operate. Under rare circumstances, this minimum flow rate may require users to change their hot water use habits and could increase hot water usage. On the other hand, accidental demand for hot water from improperly positioned single-lever faucets may result in a flow rate of less than the 0.5 gpm, thus eliminating energy usage that would occur with a storage gas unit.

Economic Impact

The economics of gas tankless water heaters in single-family new construction and in selected retrofit applications can result in a positive cash flow for the

homeowner. Preferred markets for tankless water heaters include households with high hot water use, custom homes where several storage water heaters are typically installed, and innovators and early adopters who embrace advanced technology. The retrofit market, although sizable, is complicated by the issues identified above. Although contractors interviewed in the course of our research indicated that approximately 25% to 50% of existing homes are candidates for tankless heaters, this estimate may be optimistic. In any case, if tankless water heaters achieve a significant market share, the natural gas energy savings and reduced carbon emissions would be significant. Consumers, design professionals, builders, and the plumbing trades must be educated on the benefits and special requirements of tankless gas water heaters.



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FOR MORE INFORMATION:

The full report on the study of tankless water heaters is available at http://207.67.203.54/Qelibrary4_p40007_documents/Tankless%20water%20heater.pdf.